Course learning sequence

→ **Geodesy and Earth figure**

→ **simple spherical Earth**

→ **spherical distance**

→ **simple projection**

→ **ellipsoid**

→ **projection distortion**

→ **Mercator**

→ **Cartopy visualization**

→ **UTM Thailand**

→ **Thai datum / CORS / ITRF**

→ **Web Mercator**

→ **Thai–Cambodia map case**

→ **LDP**

→ **digital cartography + digitization**

→ **final project**

Recommended standard output for every Colab exercise

Output Type	Requirement
Code	Colab notebook .ipynb
Numerical result	pandas table / CSV summary
Map output	SVG, PNG map or interactive HTML map
Short explanation	5–10 lines of technical interpretation
Survey relevance	Explain how it affects real survey / mapping work

Map Projection and Digital Cartography for Bachelor Survey Engineering

Total: 15 weeks × 3 hours/week

Format: 2 hours lecture + 1 hour Colab / in-class exercise

Week	Main Topic	Course Content, 2 hours	Colab / In-class Exercise, 1 hour
1	**Introduction to Geodesy and Earth Figure**	Meaning of geodesy; role of geodesy in surveying and mapping; physical Earth surface, geoid, ellipsoid, sphere; why Earth figure matters for map projection	Setup Colab; plot simple Earth models: sphere, ellipsoid cross-section; compare radius, semi-major axis, semi-minor axis
2	**Map Projection from Simple Sphere**	Earth as a sphere; latitude and longitude; meridian and parallel; globe-to-map concept; why projection is needed	Create world graticule; plot meridians and parallels; mark Thailand and Southeast Asia
3	**Spherical Geometry and Distance**	Great circle, small circle, central angle; spherical law of cosines; haversine formula; numerical stability for short baselines	Compute Bangkok–Chiang Mai and Bangkok–Phnom Penh distance; compare law of cosines vs haversine at 100 km, 1 km, 1 m, 1 cm
4	**Simple Cylindrical Projection**	Plate Carrée / Equirectangular projection; simple relation between longitude-latitude and x-y; projection from sphere to plane	Write Python functions for lon/lat → x/y; project graticule and GPS points; plot Thailand in simple cylindrical projection
5	**From Sphere to Spheroid / Ellipsoid**	Why Earth is not a sphere; semi-major axis, semi-minor axis, flattening; WGS84 ellipsoid; geodetic latitude vs geocentric latitude	Compare spherical distance with ellipsoidal geodesic distance using pyproj.Geod; calculate difference in meters and ppm

Week	Main Topic	Course Content, 2 hours	Colab / In-class Exercise, 1 hour
6	**Projection Distortion**	Scale distortion, distance distortion, area distortion, angular distortion, shape distortion; Tissot indicatrix concept	Build sample grid over Thailand; compute geodesic distance vs projected distance; create distortion table and simple distortion map
7	**Mercator Projection on Sphere and Ellipsoid**	Mercator concept; conformality; meridian stretching; high-latitude distortion; spherical Mercator vs ellipsoidal Mercator	Implement spherical Mercator formula manually; compare with pyproj EPSG:3857; explain differences
8	**Cartopy for Projection Visualization**	Cartopy projection classes; coastlines; gridlines; map extent; projection comparison for teaching and analysis	Create projection gallery: Plate Carrée, Mercator, Orthographic, Transverse Mercator; same extent over Thailand / Southeast Asia
9	**Transverse Mercator and UTM for Thailand**	Transverse Mercator concept; central meridian; false easting/northing; scale factor; UTM Zone 47N and 48N in Thailand	Convert sample survey points to UTM 47N / 48N; identify correct UTM zone from longitude; compare distance between zones
10	**Thai Indian Datum 1975, NCDC-CORS / ITRF, and Transformation**	Local datum vs global datum; Indian 1975; WGS84; ITRF realization and epoch; CORS coordinate framework; transformation workflow	Inspect CRS definitions and PROJ strings; transform sample points conceptually between Indian 1975, WGS84, and ITRF; document transformation parameters
11	**Pseudo-Mercator / Web Mercator**	EPSG:3857; web map tiles; Google Maps / OSM display CRS; why Web Mercator is	Overlay points and polygons on OSM basemap using contextily or folium; compare

Week	Main Topic	Course Content, 2 hours	Colab / In-class Exercise, 1 hour
		useful for visualization but poor for survey measurement	Web Mercator distance with geodesic distance
12	**Thai–Cambodia Border Map Projection Case Study**	1:50,000 and 1:200,000 topographic maps; projection and grid issues; border mapping; scale and map distance; UTM zone consideration near border	Prepare Thai–Cambodia border sample line; compute ground distance, map distance at 1:50,000 and 1:200,000; compare UTM and Web Mercator
13	**Low Distortion Projection, LDP**	Engineering project projection; ground-to-grid distortion; combined scale factor; project-centered CRS; when LDP is better than UTM	Design local engineering CRS around project center; compute grid-to-ground distortion; compare LDP with UTM
14	**Digital Cartography, Basemap Integration, Map Transformation and Digitization**	Map scale, symbolization, color, label, legend, north arrow, layout; basemap limitations; scanned map georeferencing; similarity, affine, polynomial, TPS; digitizing old maps	Create thematic map with basemap; fit similarity/affine transformation from control points; digitize/georeference sample map features
15	**Final Digital Mapping Project and Presentation**	CRS selection for web map, GIS, cadastral map, engineering survey, and national mapping; course synthesis; projection, datum, distortion, LDP, cartographic output	Student presentation; submit Colab notebook, final map, distortion analysis, and short technical report